

ADAR'S STUDY SERIES

LINEAR ALGEBRA

PART 3 · VECTORS, LINEAR COMBINATIONS & SPAN

Vector Operations · Linear Combinations · Span · Matrix-Vector Products · Existence of Solutions

Deep Violet & Amber Edition

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1 · OVERVIEW & GOALS

- Learn what vectors are, and what $\mathbb{R}^n$ represents.
- Define a linear combination and the span of a set of vectors.
- Understand how to determine whether a vector $\vec{b}$ can be written as a linear combination of a set of vectors — both computationally and conceptually.

2 · WHAT IS $\mathbb{R}^n$?

$\mathbb{R}$ — Real Numbers (1-Dimensional)

Represents the real number line.



$\mathbb{R}^2$ — The Plane

$$\mathbb{R}^2 = \{ \text{all } (x, y) \text{ such that } x, y \in \mathbb{R} \}$$

(Set of ordered pairs)

$\mathbb{R}^3$ — 3D Space

$$\mathbb{R}^3 = \{ \text{all } (x, y, z) \text{ such that } x, y, z \in \mathbb{R} \}$$

(Set of ordered triples)

General Definition — $\mathbb{R}^n$

$$\mathbb{R}^n := \text{Set of ordered } n\text{-tuples } (x_1, x_2, \dots, x_n) \text{ of reals}$$

(where "!=" means "defined to be equal to")

Note: Elements of $\mathbb{R}^n$ are usually written as a column vector:

$\mathbb{R}^n = \text{set of all such columns, where } x_1, \dots, x_n \in \mathbb{R}$

$$\mathbb{R}^n = \begin{bmatrix} x_1 \\ x_2 \\ : \\ x_n \end{bmatrix} \quad \text{where } (x_1, x_2, \dots, x_n) \in \mathbb{R}$$

3 · VECTOR NOTATION & THE ZERO VECTOR

General Vector

We write a general vector in $\mathbb{R}^n$ as:

$$\vec{x} = \begin{bmatrix} x_1 \\ : \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

(The arrow above x represents that it is a vector.)

The Zero Vector

$$\vec{0} = \begin{bmatrix} 0 \\ 0 \\ : \\ 0 \end{bmatrix} \in \mathbb{R}^n$$

(The individual values inside a vector are called entries of the vector.)

4 · VECTOR OPERATIONS

We can add/subtract and scale vectors.

Let $\vec{x} = [x_1, \dots, x_n]$, $\vec{y} = [y_1, \dots, y_n] \in \mathbb{R}^n$, and let $c \in \mathbb{R}$ be a scalar (constant).

Vector Addition

$$\vec{x} + \vec{y} := \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

Vector Subtraction

$$\vec{x} - \vec{y} := \begin{bmatrix} x_1 - y_1 \\ x_2 - y_2 \\ \vdots \\ x_n - y_n \end{bmatrix}$$

Scalar Multiplication (Scaling)

$$c\vec{x} = \begin{bmatrix} c \cdot x_1 \\ \vdots \\ c \cdot x_n \end{bmatrix}$$

$$(Scalar) \cdot (Vector) = Vector$$

5 · GEOMETRIC INTERPRETATION IN $\mathbb{R}^2$

Convention: Always start vectors at the origin $(0, 0)$.

Example 1 — Vector Addition

Let $\vec{x} = [2, 1]$ and $\vec{y} = [1, 3] \in \mathbb{R}^2$.

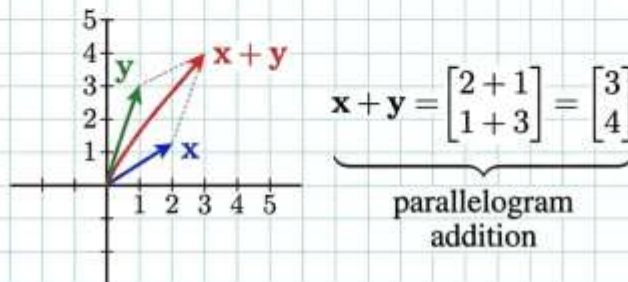
$$\vec{x} + \vec{y} = [2+1, 1+3] = [3, 4]$$

Parallelogram Rule for Addition

Picture in $\mathbb{R}^2$

Ex 1 Let $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, in $\mathbb{R}^2$. Convention to start vector at origin.

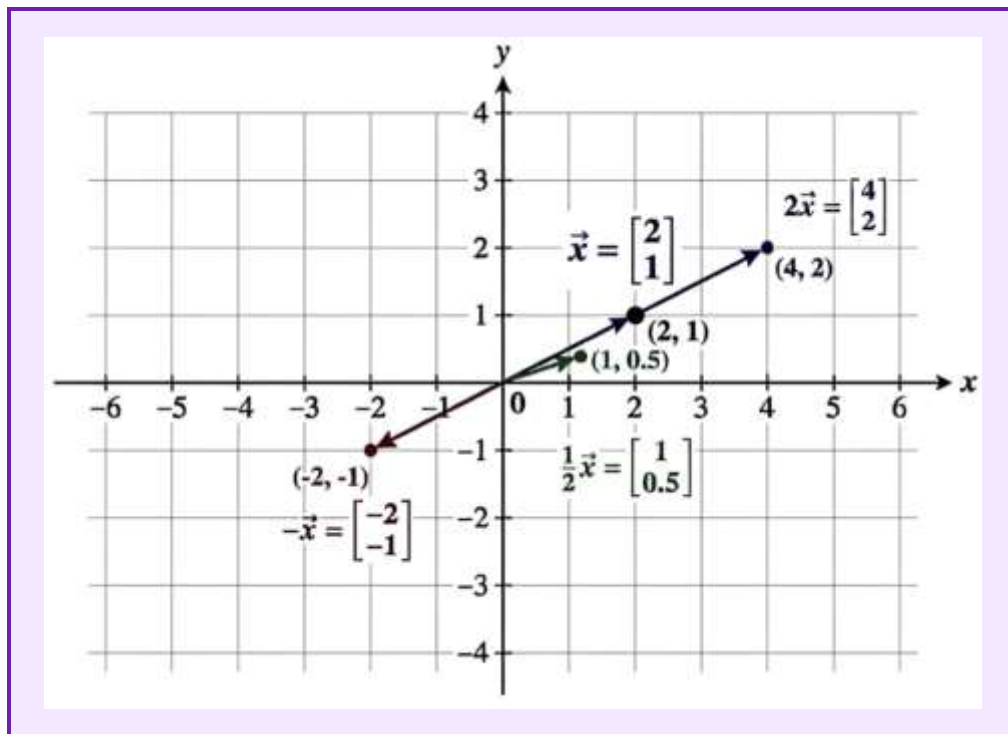
(a) Draw $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{x} + \mathbf{y}$ in xy -plane.



Example 2 — Scalar Scaling

Let $\vec{x} = [2, 1]$. We sketch $\vec{x}$, $2\vec{x}$, $\frac{1}{2}\vec{x}$, and $-\vec{x}$:

- $2\vec{x} = [4, 2]$
- $\frac{1}{2}\vec{x} = [1, 0.5]$
- $-\vec{x} = [-2, -1]$



6 · BASIC ARITHMETIC PROPERTIES OF VECTORS

Vectors in $\mathbb{R}^n$ satisfy fundamental algebraic properties for any vectors $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$ and scalar $c \in \mathbb{R}$

Associativity of Addition

$$(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$$

Additive Identity

$$\vec{0} + \vec{v} = \vec{v}$$

Zero Scalar Multiplication

$$\mathbf{0} \cdot \vec{v} = \vec{0}$$

Distributivity

$$c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$$

7 · LINEAR COMBINATION & SPAN

Linear Combination

Definition: Let $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k$ be vectors in $\mathbb{R}^n$. Let $c_1, c_2, \dots, c_k \in \mathbb{R}$ be scalars (called weights). Then the expression:

$c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_k\vec{v}_k$ is called a linear combination of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k$.

Note: A linear combination of vectors results in another vector.

Span

Definition: The span of a set of vectors is the set of all possible linear combinations of those vectors:

$\text{Span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\} = \text{All linear combinations of } \vec{v}_1, \dots, \vec{v}_k$

Conceptual Meaning

$\text{Span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\} = \text{Everywhere we can get to in } \mathbb{R}^n \text{ only traveling in direction of } \vec{v}_1, \dots, \vec{v}_k$

Key Property

The zero vector $\vec{0}$ is always in $\text{Span}\{\vec{v}_1, \dots, \vec{v}_k\}$.

Why? By choosing all weights to be 0, we get:

$$0\vec{v}_1 + 0\vec{v}_2 + \dots + 0\vec{v}_k = \vec{0}$$

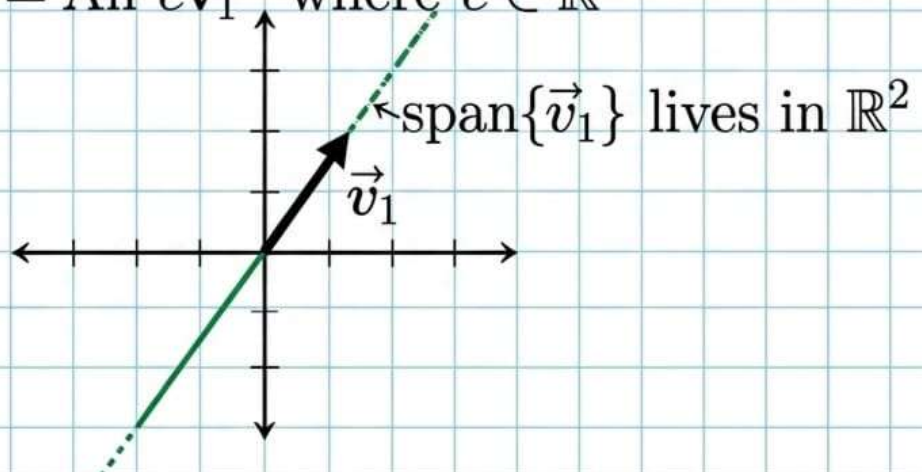
8 · GEOMETRIC EXAMPLES OF SPAN

Example 2(a) — Single Vector in $\mathbb{R}^2$

Ex 2 Describe the span of the vectors geometrically.

(a) Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in $\mathbb{R}^2$.

$\text{Span}\{\vec{v}_1\} = \text{All } c\vec{v}_1 \text{ where } c \in \mathbb{R}$



Example 2(b) — Two Linearly Independent Vectors in $\mathbb{R}^3$

Let $\vec{v}_1 = [1, 0, 0]$ and $\vec{v}_2 = [0, 1, 0]$ in $\mathbb{R}^3$.

$$c_1\vec{v}_1 + c_2\vec{v}_2 = c_1[1, 0, 0] + c_2[0, 1, 0] = [c_1, c_2, 0]$$

Geometric Interpretation: The xy-plane in 3D-space ($\mathbb{R}^3$).

Important Note: $\text{Span}\{\vec{v}_1, \vec{v}_2\} \neq \mathbb{R}^2$ because the vectors in this span have 3 components and live in $\mathbb{R}^3$, whereas elements of $\mathbb{R}^2$ only have 2 components.

Example 2(c) — Two Parallel (Dependent) Vectors in $\mathbb{R}^3$

Let $\vec{v}_1 = [1, 3, -4]$ and $\vec{v}_2 = [\frac{1}{4}, \frac{3}{4}, -1]$ in $\mathbb{R}^3$.

Observe that $\vec{v}_1 = 4\vec{v}_2$. Substituting this into a general linear combination gives:

$$c_1\vec{v}_1 + c_2\vec{v}_2 = c_1(4\vec{v}_2) + c_2\vec{v}_2 = (4c_1 + c_2)\vec{v}_2$$

Since $(4c_1 + c_2)$ is just a scalar constant:

$$\text{Span}\{\vec{v}_1, \vec{v}_2\} = \text{Span}\{\vec{v}_2\} = \text{Line through origin and } [\frac{1}{4}, \frac{3}{4}, -1] \text{ in } \mathbb{R}^3$$

9 · MATRIX REPRESENTATION OF A LINEAR SYSTEM

Suppose a linear system has m equations and n variables. Its augmented matrix is expressed as:

$$\begin{aligned} & \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ : & : & & : & : \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right] \\ &= \left[\begin{array}{cccc|c} \vec{a}_1 & \vec{a}_2 & \dots & \vec{a}_n & \vec{b} \end{array} \right] \\ &= \left[A \mid \vec{b} \right] \end{aligned}$$

- $A = [\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n]$ is the $m \times n$ coefficient matrix.
- Each column $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ is a column vector in $\mathbb{R}^m$.
- $\vec{b} = [b_1, b_2, \dots, b_m]$ is the right-hand side constant vector in $\mathbb{R}^m$.

10 · EQUIVALENCES FOR SYSTEM SOLUTIONS

Let $\vec{x} = [x_1, x_2, \dots, x_n] \in \mathbb{R}^n$ be a candidate solution vector for $[A \mid \vec{b}]$. The following statements are equivalent:

1. System of Equations Representation

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

2. Vector Equation Representation

$$\begin{array}{ccccccc} [a_{11}] & [a_{12}] & & [a_{1n}] & [b_1] \\ x_1 \cdot [a_{21}] + x_2 \cdot [a_{22}] + \dots + x_n \cdot [a_{2n}] & = & [b_2] \\ [\vdots] & [\vdots] & & [\vdots] & [\vdots] \\ [a_{m1}] & [a_{m2}] & & [a_{mn}] & [b_m] \end{array}$$

$$x_1\vec{a}_1 + x_2\vec{a}_2 + \dots + x_n\vec{a}_n = \vec{b}$$

3. Linear Combination / Span Interpretation

$\vec{b}$ is a linear combination of column vectors $\vec{a}_1, \dots, \vec{a}_n$.

$$\text{Equivalently: } \vec{b} \in \text{Span}\{\vec{a}_1, \dots, \vec{a}_n\}$$

11 · THE MATRIX-VECTOR PRODUCT $A\vec{x}$

Definition: If $A = [\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n]$ is an $m \times n$ matrix (with m rows and n columns, where $\vec{a}_i \in \mathbb{R}^m$) and $\vec{x} = [x_1, x_2, \dots, x_n] \in \mathbb{R}^n$, then the product $A\vec{x}$ is defined as:

$$A\vec{x} := x_1\vec{a}_1 + x_2\vec{a}_2 + \dots + x_n\vec{a}_n$$

This product $A\vec{x}$ represents a linear combination of the columns of A with weights given by the entries of $\vec{x}$, resulting in a vector in $\mathbb{R}^m$.

12 · FUNDAMENTAL THEOREM — EXISTENCE OF SOLUTIONS

The following statements are equivalent:

- The linear system $[A \mid \vec{b}]$ has a solution
- $\Leftrightarrow \vec{b}$ is a linear combination of $\vec{a}_1, \dots, \vec{a}_n$ (same as saying $\vec{b} \in \text{span}\{\vec{a}_1, \dots, \vec{a}_n\}$)
- $\Leftrightarrow A\vec{x} = \vec{b}$ for some $\vec{x} \in \mathbb{R}^n$
- $\Leftrightarrow$ The echelon form of $[A \mid \vec{b}]$ has NO row of the form $[0 \ \dots \ 0 \mid \blacksquare]$ with $\blacksquare \neq 0$

13 · WORKED EXAMPLE — TESTING SPAN MEMBERSHIP

Example 3

Question:

Is $[1, 2, 3] \in \text{Span}\{[-1, 1, 2], [2, 1, 0], [1, 2, 2]\}$?
(i.e., is $[1, 2, 3]$ a linear combination of those vectors?)

Solution Procedure

Check if the linear system given by the augmented matrix is consistent:

Initial Matrix

$$\begin{bmatrix} -1 & 2 & 1 & | & 1 \\ 1 & 1 & 2 & | & 2 \\ 2 & 0 & 2 & | & 3 \end{bmatrix}$$

Apply row operations: $R_1 + R_2 \rightarrow R_2$, $2R_1 + R_3 \rightarrow R_3$

$$\begin{bmatrix} -1 & 2 & 1 & | & 1 \\ 0 & 3 & 3 & | & 3 \\ 0 & 4 & 4 & | & 5 \end{bmatrix}$$

Simplify rows: $(-1) R_1 \rightarrow R_1$, $(1/3) R_2 \rightarrow R_2$

$$\begin{bmatrix} 1 & -2 & -1 & | & -1 \\ 0 & 1 & 1 & | & 1 \\ 0 & 4 & 4 & | & 5 \end{bmatrix}$$

Eliminate Row 3: $-4R_2 + R_3 \rightarrow R_3$

$$\begin{bmatrix} 1 & -2 & -1 & | & -1 \\ 0 & 1 & 1 & | & 1 \\ 0 & 0 & 0 & | & 1 \end{bmatrix}$$

Conclusion

The row $[0 \ 0 \ 0 \ | \ 1]$ indicates that the system is inconsistent.

$[1, 2, 3]$ is NOT in $\text{Span}\{ [-1, 1, 2], [2, 1, 0], [1, 2, 2] \}$

14 · WORKED EXAMPLE — FINDING WEIGHTS IN A SPAN

Example 4

Question:

$$\text{Is } \vec{b} = [-9, 3, 10] \in \text{Span}\{ [-1, 1, 2], [2, 1, 0] \} ?$$

(If yes, find weights x_1, x_2 such that $\vec{b} = x_1 \vec{v}_1 + x_2 \vec{v}_2$)

Solution Procedure

Set up the augmented matrix:

$$\begin{bmatrix} -1 & 2 & | & -9 \\ 1 & 1 & | & 3 \\ 2 & 0 & | & 10 \end{bmatrix}$$

Row operations to Echelon Form: $R_1 + R_2 \rightarrow R_2$, $2R_1 + R_3 \rightarrow R_3$

$$\begin{bmatrix} -1 & 2 & | & -9 \\ 0 & 3 & | & -6 \\ 0 & 4 & | & -8 \end{bmatrix}$$

Rescale rows: $R_1(-1) \rightarrow R_1$, $R_2(1/3) \rightarrow R_2$, $R_3(1/4) \rightarrow R_3$

$$\begin{bmatrix} 1 & -2 & | & 9 \\ 0 & 1 & | & -2 \\ 0 & 1 & | & -2 \end{bmatrix}$$

Eliminate Row 3: $R_2 - R_3 \rightarrow R_3$

Echelon Form (EF)

$$\begin{bmatrix} 1 & -2 & | & 9 \\ 0 & 1 & | & -2 \\ 0 & 0 & | & 0 \end{bmatrix}$$

Observation: The Echelon Form has no row of the form $[0 \dots 0 \mid \blacksquare]$ where $\blacksquare \neq 0$. Thus, a solution exists!

Back Substitution / RREF: $2R_2 + R_1 \rightarrow R_1$

Reduced Row Echelon Form (RREF)

$$\begin{bmatrix} 1 & 0 & | & 5 \\ 0 & 1 & | & -2 \\ 0 & 0 & | & 0 \end{bmatrix}$$

Conclusion

$$\begin{aligned} x_1 &= 5 \\ x_2 &= -2 \end{aligned}$$

Thus:

$$[-9, 3, 10] = 5 \cdot [-1, 1, 2] - 2 \cdot [2, 1, 0]$$

15 • FUNDAMENTAL THEOREM — SPANNING THE ENTIRE SPACE $\mathbb{R}^m$

Note: Key Equivalent Conditions

Let A be an $m \times n$ matrix with columns $\vec{a}_1, \dots, \vec{a}_n$. The following statements are equivalent ($\Leftrightarrow$):

1. Linear system $[A \mid \vec{b}]$ has a solution for every $\vec{b} \in \mathbb{R}^m$.
2. $\text{Span}\{\vec{a}_1, \dots, \vec{a}_n\} = \mathbb{R}^m$
3. The reduced matrix never has a row of the form $[0 \dots 0 \mid \blacksquare]$ (where $\blacksquare \neq 0$).
4. The coefficient matrix A has a pivot in every row.

16 · WORKED EXAMPLE — TESTING FULL SPACE SPAN

Example 5

Question:

Do vectors $[1,2,3]$, $[1,0,1]$, $[2,-1,1]$ span all of $\mathbb{R}^3$?

Solution Procedure

Form the matrix A and check if there is a pivot in each row:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & -1 \\ 3 & 1 & 1 \end{bmatrix}$$

Row Reduction

Initial Coefficient Matrix:

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & -1 \\ 3 & 1 & 1 \end{bmatrix}$$

Eliminate lower entries in Column 1: $-2R_1 + R_2 \rightarrow R_2$, $-3R_1 + R_3 \rightarrow R_3$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -5 \\ 0 & -2 & -5 \end{bmatrix}$$

Eliminate Row 3: $R_3 - R_2 \rightarrow R_3$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

Conclusion

Since row 3 contains all zeros, not every row has a pivot (there are only 2 pivots, but 3 rows are required).

The columns do NOT span $\mathbb{R}^3$.

17 · KEY POINTS & SUMMARY

- $\mathbb{R}^n$ is the set of ordered n-tuples, usually written as column vectors.
- Vectors can be added, subtracted, and scaled entry wise.
- Vector addition follows the parallelogram rule geometrically.
- A linear combination $c_1\vec{v}_1 + \dots + c_k\vec{v}_k$ always produces another vector.
- $\text{Span}\{\vec{v}_1, \dots, \vec{v}_k\}$ is the set of ALL linear combinations — and always contains $\vec{0}$.
- A linear system $[A \mid \vec{b}]$ has a solution $\Leftrightarrow \vec{b} \in \text{Span}\{\text{columns of } A\}$
 $\Leftrightarrow A\vec{x} = \vec{b}$ for some $\vec{x}$.
- A pivot-free row in the constants column of the echelon form guarantees a solution exists.
- Columns of A span the ENTIRE space $\mathbb{R}^m$ only when A has a pivot in every row.

END OF PART 3 · NEXT: PART 4 — MATRIX EQUATION & PARAMETRIC VECTOR FORM